

The Digits Are Not the Number

This document was generated by a large language model (LLM) and may contain errors.

1 A picture and its immediate failure

Ask someone what an irrational number is and you will often be told: a number whose decimal expansion goes on forever. The picture behind the answer is a tape of digits unrolling to the right without end, never settling down.

The picture is not so much wrong as uninformative, and it is worth being precise about why. Consider

$$\frac{1}{3} = 0.333\dots, \quad \frac{1}{2} = 0.5000\dots, \quad \sqrt{2} = 1.41421356\dots$$

All three expansions are infinite. Under the base-10 algorithm every real number receives an infinite string of digits; terminating expansions are those that terminate in an infinite tail of zeros. So infinitude of digits is a property of the algorithm, shared by every real, and it separates nothing.

What actually separates the cases is periodicity.

Proposition 1.1. *A real number x is rational if and only if its base- b expansion is eventually periodic, for any (equivalently, every) integer base $b \geq 2$.*

The proof in one direction is long division: computing the expansion of p/q produces at each step a remainder in $\{0, 1, \dots, q-1\}$, and the state of the algorithm is determined by that remainder, so within q steps some remainder recurs and the process cycles. The converse is the geometric series: an eventually periodic expansion is a finite decimal plus a scaled geometric sum with rational ratio b^{-k} , hence rational.

Proposition 1.1 is a genuine theorem, and it does exactly what the naive picture failed to do. It also has an instructive shape. Rationality is a base-independent notion — p/q is a ratio of integers regardless of how we write it — and the proposition asserts that eventual periodicity, which looks base-dependent, is in fact base-independent too. That the equivalence holds in every base is a fact requiring proof, not a triviality.

But notice what has happened. We began with a proposed definition of irrationality and ended with a characterization by a property of a representation, a property whose base-independence is itself a theorem. That inversion is the subject of this essay. The digit expansion is a coordinate system imposed on the real line, and like any coordinate system it has features of its own that do not belong to the object it describes. Below I argue that the decimal representation is in three specific respects a poor coordinate system; that better ones exist; that the difference between them is not cosmetic but reveals structure invisible from the digit side; and that pushing on the question of what a representation must supply leads, fairly quickly, to questions about what it means for a real number to exist at all.

2 Three defects

2.1 Base dependence

The number $1/3$ has the base-10 expansion $0.333\dots$ and the base-3 expansion 0.1 . The number $1/2$ terminates in base 10 and repeats in base 3, where it is $0.111\dots$. Nothing about the number changes; the visible texture of its representation changes completely.

Termination, then, is not a property of a real number. It is a property of a pair: a number and a base. A rational p/q in lowest terms terminates in base b exactly when every prime dividing q divides b . The distinction the decimal system makes most salient — between 0.5 , which looks finite and tame, and $0.333\dots$, which looks infinite and unruly — is an artifact of the factorization of 10.

Proposition 1.1 shows that one can recover a base-independent notion (rationality) from a base-dependent formalism (periodicity), but only by proving that the base does not matter. Any statement about digits carries this burden. Until the burden is discharged, a digit-level observation is a statement about notation.

2.2 Non-uniqueness

The map from digit strings to real numbers is not injective:

$$0.999\dots = 1.000\dots$$

The identity is forced by the definition of the expansion as the sum of a series: $\sum_{n \geq 1} 9 \cdot 10^{-n} = 1$. Every terminating expansion has a second representation ending in repeated $(b-1)$'s, and these are the only collisions.

The consequence for our purposes is structural rather than arithmetical. If a real number simply *were* its digit string, then $0.999\dots$ and 1 would be different numbers, which they are not. To make the identification work one must first quotient the space of digit strings by the relation just described. The corrected slogan — a real number is an equivalence class of digit strings — is already conceding the point: the digit string is not the number, because knowing the number does not determine the string.

I will not belabor this. It is the best-known item in this catalogue and the shallowest.

2.3 Carrying, or: the representation is arithmetically inert

The serious defect is that one cannot compute with digit expansions.

Consider adding $x = 0.333\dots$ and $y = 0.666\dots$. Suppose an adversary hands you the digits one at a time and asks for the first digit of $x + y$ after you have seen any finite number of them. After a million 3's and a million 6's the sum of the visible parts is $0.999\dots 9$, and you must announce either 0 or 1 as the leading digit of the result. You cannot. If the expansions continue as 3's and 6's forever, the sum is exactly 1 and the answer is 1; if the very next digit of x is 2 instead of 3, the sum begins 0.9 and the answer is 0. No finite prefix of the inputs determines even the first digit of the output.

The phenomenon is unbounded carry propagation, and it is not an edge case cleverly constructed for the occasion. It is generic near any point with two expansions, that is, near every terminating rational, a dense set. State the consequence precisely:

Proposition 2.1. *Addition is not computable as a function from pairs of digit streams to digit*

streams. There is no algorithm which, reading the digits of x and y in order, emits the digits of $x + y$ in order, with each emitted digit depending on only finitely many read digits.

This is a strong statement and it disqualifies the representation for any purpose involving actual computation. The field operations, the operations that make $\mathbb{R}$ what it is, do not act on digit streams in any local way. A representation on which one cannot add is not a representation of a field; it is a labelling scheme.

That the defect was recognized early is worth recording. Turing’s 1936 paper was called “On Computable Numbers,” and he defined a computable real as one whose decimal expansion is produced by a machine. The correction he published the following year was concerned precisely with the inadequacy of that choice. Modern computable analysis does not use digit streams. It represents a real by a sequence of nested rational intervals of shrinking width, or by a Cauchy sequence of rationals together with an explicit modulus of convergence:

$$(q_n)_{n \in \mathbb{N}} \subset \mathbb{Q} \quad \text{with} \quad |q_n - q_m| \leq 2^{-n} \quad \text{for } m \geq n.$$

On these representations the field operations are computable, because an interval enclosing x and an interval enclosing y combine to give an interval enclosing $x + y$, and the enclosures can be made as tight as desired by going deeper into the inputs. The information is the same; its organization is not.

Here is a test for the naive picture, worth carrying around. What is the digit expansion of

$$\sqrt{2} + (2 - \sqrt{2})?$$

The answer is 2, and the reason is the field axioms. No digit-level procedure reaches it: both summands have expansions that are, so far as anyone can prove, statistically structureless, and the cancellation that makes the answer exact is invisible at every finite stage. The gap between what you know and what the digits show you is the measure of how much the representation is not the number.

3 What the object is

If the digits are a chart, what is being charted?

There are several standard constructions and they agree. Dedekind’s takes a real number to be a cut: a partition of $\mathbb{Q}$ into a downward-closed set A with no greatest element and its complement B . The irrationals are the cuts whose associated A has no rational supremum; $\sqrt{2}$ is the cut $A = \{q \in \mathbb{Q} : q < 0 \text{ or } q^2 < 2\}$, and the construction says nothing about digits. Cantor’s takes a real to be an equivalence class of Cauchy sequences of rationals under the relation $(a_n) \sim (b_n)$ iff $a_n - b_n \rightarrow 0$; the irrationals are the classes containing no constant sequence.

The axiomatic view subsumes both.

Theorem 3.1. *There is, up to isomorphism of ordered fields, exactly one complete ordered field.*

$\mathbb{R}$ is that field. Both constructions produce it; both are therefore descriptions of the same object, and neither is more the real numbers than the other. What matters for the present argument is the negative observation: no formulation of the definition mentions digits, or bases, or expansions. The decimal system is an afterthought, a convenience introduced once the object is in hand, and every property one might read off a digit string must be reconnected to the definition before it counts as a property of a number.

4 A better representation

Criticism of a representation is cheap unless one can exhibit an alternative. The continued fraction expansion is the alternative, and comparing the two converts a complaint into an argument.

Every real x has an expansion

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}, \quad a_0 \in \mathbb{Z}, \quad a_i \in \mathbb{N}_{\geq 1} \ (i \geq 1),$$

written $x = [a_0; a_1, a_2, a_3, \dots]$. It is produced by the obvious algorithm: take the integer part, invert the remainder, repeat. For irrational x the expansion is infinite and unique.

Four points of comparison.

It is base-free. The algorithm involves no choice of b . The first defect vanishes because there is nothing for it to attach to.

The rational/irrational split is termination, not periodicity. The algorithm applied to p/q is the Euclidean algorithm and halts; applied to an irrational it does not. So the dichotomy that decimals express through the subtle condition “eventually periodic” — a condition requiring Proposition 1.1 to connect it to anything — appears here as the crudest possible one.

Periodicity is then free to mean something. It does.

Theorem 4.1 (Lagrange). *An irrational x has an eventually periodic continued fraction expansion if and only if x is a quadratic irrational, that is, a root of a quadratic with integer coefficients.*

Thus $\sqrt{2} = [1; 2, 2, 2, \dots]$ and $\frac{1+\sqrt{5}}{2} = [1; 1, 1, 1, \dots]$. In the decimal system, periodicity is spent on distinguishing $1/7$ from $1/2$, a distinction of no structural interest; here it identifies the first nontrivial layer of the algebraic hierarchy. The same formal feature, allocated to a different representation, detects a different and much better thing.

The truncations are optimal. The convergents p_n/q_n obtained by cutting the expansion at a_n satisfy

$$\left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2},$$

and, more to the point, they are best approximations in a strong sense: if $q \leq q_n$ and $p/q \neq p_n/q_n$ then $|qx - p| > |q_n x - p_n|$. Every exceptionally good rational approximation to x is a convergent. Decimal truncation offers nothing comparable; 3.14159 approximates π to six figures with denominator 10^5 , while the convergent 355/113 does better with denominator 113.

The comparison establishes the essay’s thesis in its strong form. The two representations carry the same information — each determines the number, hence the other — and yet one makes the arithmetic of approximation legible and the other conceals it. Legibility is not a property of the number. It is a property of the chart.

5 What the good chart reveals

Continued fractions do more than tidy up. They open onto a quantitative theory of irrationality that the digit picture cannot see at all.

The question is: how well can a given irrational be approximated by rationals with small denominators? Define the irrationality measure $\mu(x)$ as the infimum of exponents μ such that

$$\left| x - \frac{p}{q} \right| < \frac{1}{q^\mu}$$

has only finitely many rational solutions p/q . This is base-independent by construction, and it stratifies the irrationals.

Dirichlet's theorem gives $\mu(x) \geq 2$ for every irrational: the inequality with exponent 2 always has infinitely many solutions, the convergents among them. At the other end, Liouville showed that an algebraic number of degree d cannot be approximated to order better than d , and used this to construct the first numbers proved transcendental — the point being that a number approximated *too* well by rationals cannot be algebraic. Liouville's constant

$$L = \sum_{k=1}^{\infty} 10^{-k!} = 0.110001000000000000000001\dots$$

has $\mu(L) = \infty$. Roth's theorem (1955) closed the gap at the top: every irrational algebraic number has $\mu = 2$ exactly, the minimum. Algebraic irrationals are, uniformly and regardless of degree, as badly approximable as an irrational is permitted to be.

So there is a real hierarchy — rational, quadratic irrational, higher algebraic, transcendental, Liouville — graded by approximability, and it is detected by the growth of the partial quotients a_n : bounded partial quotients mean badly approximable, huge partial quotients mean the convergents are exceptionally good. The golden ratio, all of whose partial quotients equal 1, is the worst-approximable number there is, and this is the precise sense in which it is the “most irrational.” None of this structure is visible in a decimal expansion.

Fairness requires one concession. L was written above as a decimal, and its structure is entirely obvious there — a sparse pattern of 1's at positions $k!$. Liouville numbers are visible in the digit picture precisely because extremely good rational approximation corresponds to extremely long runs of zeros. The decimal representation is not uniformly blind; it is blind in a specific direction. It sees sparseness and it cannot see approximation quality in general.

6 Almost every number, and no number in particular

Push the digit picture in the one direction where it seems to promise a theory — the statistics of the digits themselves — and it produces a result that is worth examining closely, because the shape of its failure is instructive.

Definition 6.1. x is normal in base b if for every k , each of the b^k possible blocks of k digits occurs in the base- b expansion of x with asymptotic frequency b^{-k} . It is absolutely normal if it is normal in every base $b \geq 2$.

Theorem 6.2 (Borel, 1909). *Almost every real number, with respect to Lebesgue measure, is absolutely normal.*

The exceptional set has measure zero. And yet: it is not known whether π is normal in any base. Nor e , nor $\sqrt{2}$, nor $\log 2$, nor any constant that arose from mathematics rather than being constructed for this purpose. Not one. The numbers we can prove normal are the ones built to be normal, such as Champernowne's constant

$$C_{10} = 0.123456789101112131415\dots,$$

formed by concatenating the integers, which is normal in base 10 by construction and was published as the first explicit example.

Two lessons. The first is about the digit picture: it supports a statistical theory that proves almost everything about almost all numbers and almost nothing about any number one can name. Whatever is doing the work in Borel's theorem is measure theory, not the expansion; the expansion contributes the statement and no leverage on it.

The second lesson is the more disquieting. "Almost every real is normal, and no named real is known to be" is an instance of a pattern. The named reals form a countable set — there are only countably many finite descriptions — and a countable set has measure zero. The typical real, in the measure-theoretic sense, is one we can never name, discuss, or single out. Every real number that will ever appear in a paper lies in a null set. The continuum is mostly inaccessible, and this is not a limitation of technique but a cardinality fact.

Which raises the question of what such numbers are, and whether a representation should be required to hand us anything at all.

7 Existence, and the point of a representation

Turing's diagonal argument settles the accounting. There are countably many Turing machines and uncountably many reals; therefore all but countably many reals are uncomputable, in the sense that no algorithm produces arbitrarily good rational approximations to them. Almost every real, again in the measure sense, is a number whose digits — or convergents, or nested intervals; the representation is irrelevant here — cannot be generated by any procedure whatsoever.

The completeness of $\mathbb{R}$, which is the axiom that distinguishes it from $\mathbb{Q}$ and the engine of every limit argument in analysis, is what manufactures these objects, and it does so non-effectively.

Theorem 7.1 (Specker). *There is a computable, strictly increasing, bounded sequence of rationals whose limit is not a computable real number.*

The sequence is explicit: enumerate the halting Turing machines and sum 2^{-n} over those that halt. The partial sums are computable and increase to a bound; the limit encodes the halting problem. Monotone convergence, which every analysis course treats as elementary, here converts a completely effective input into a non-effective output. The supremum exists by completeness and cannot be computed by anything.

One may accept this and move on, and classical mathematics does. But one may also ask what it means to assert the existence of an object that no procedure can approach, and the constructive tradition takes the question seriously. If a real number is required to come equipped with a procedure — if to have a real number is to have a way of producing rational approximations to any specified accuracy — then the classical picture changes in ways that are not adjustments at the margin.

Trichotomy fails. Given a constructive real x one cannot in general decide which of $x < 0$, $x = 0$, $x > 0$ holds: deciding it would require knowing whether all the approximations that have not yet been computed will remain within a shrinking interval around zero, and that is not finitely checkable. Deciding it in general would settle the halting problem. The trichotomy law, which looks like a triviality about the ordering of the line, turns out to encode an assumption about what we are allowed to know.

Bishop's development of constructive analysis shows how much survives, which is a surprising amount: continuous functions on closed intervals, differentiation, integration, the essential machinery. What is lost is the freedom to case-split on undecidable comparisons, and a handful of theorems that were secretly using that freedom.

The point is not to endorse this position. The point is that the naive picture we started with — a real number is a tape of digits — is not a neutral placeholder awaiting refinement. It is a specific and contestable ontological commitment. It says that a number is a completed infinite object, given all at once, whether or not anything could ever produce it. That is exactly the commitment the constructivist declines, and exactly the one that generates Specker’s sequence and the uncountable ocean of numbers nobody will ever name. The tape of digits is not a chart of the reals. It is a chart of a particular answer to the question of what a real number is.

8 Why the bad chart survives

Given all this, one should ask why the digit representation persists, and the answer is that it is genuinely good at one thing.

Cantor’s diagonal argument requires a representation in which the numbers are laid out as sequences of independently specifiable symbols from a finite alphabet, so that a new sequence can be constructed by disagreeing with the n -th listed sequence in the n -th place. Decimal expansions provide this. The proof of the uncountability of $\mathbb{R}$ — the fact underlying every observation above about how much of the line is inaccessible — is a proof about digit strings, and it needs no more from them than that they exist and nearly determine the number. The $0.999\dots$ ambiguity intrudes and is disposed of by avoiding the digits 0 and 9 in the constructed number.

So the representation that cannot support addition is the one that establishes the cardinality of the continuum. This is not paradoxical. Digit expansions encode the reals as an infinite product of finite sets, and infinite products of finite sets are exactly what combinatorial and set-theoretic arguments handle well. What such an encoding discards is the field structure and the metric structure — which is to say, everything analysis is about. Use it for cardinality and for measure-theoretic statements about typical numbers, where it is the right tool. Do not use it for arithmetic, for approximation, or as a definition.

9 Conclusion

Is it fair to think of irrational numbers as infinite strings of digits after the dot? As a mnemonic, and with periodicity supplied in place of infinitude, it is serviceable. As an account of what an irrational number is, it fails, and the manner of the failure is more interesting than the fact.

It fails first on its own terms, being base-dependent, non-injective, and incapable of supporting addition. It fails by comparison, since the continued fraction expansion carries the same information while making visible a whole theory — approximability, the algebraic hierarchy, Lagrange’s characterization of quadratic irrationals — that the decimal expansion conceals. And it fails philosophically, in that it presupposes an answer to a live question, treating a real number as a completed infinite object handed over in full when the great majority of such objects can never be handed over at all.

The corrective is not a better representation. It is the recognition that a representation is a choice, that each choice illuminates the structure it was built to reflect and hides the rest, and that the number is the thing that persists across all of them. $\sqrt{2}$ is the positive root of $x^2 = 2$: the cut in $\mathbb{Q}$ at that point, the length of the diagonal of the unit square, the unique real whose square is 2. It is $1.41421356\dots$ and it is $[1; 2, 2, 2, \dots]$, and neither string is the number. They are shadows cast by particular projections, and the mathematics is in the object, not in the shadows.